EVAN ZHANG

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EDUCATION

University of Pennsylvania, *Jerome Fisher Program in Management and Technology* **Aug 2026 – May 2030**
- *School of Engineering and Applied Science:* BSE in Electrical Engineering, MSE in Mechanical Engineering
- *The Wharton School:* BS in Economics, Concentration in Finance

The Harker School, San Jose, CA **Aug 2022 – May 2026**
- **Weighted GPA:** 4.35/4.0 | **ACT:** 36 | **Courses:** Multivariable Calculus, Linear Algebra, AP Physics C, Discrete Math

EXPERIENCE

***Mechanical and Electrical Engineering Intern*, Vero** **Sept 2026 – Present**
- Designed a 30 × 50 × 70 in. battery energy storage system (BESS) in SolidWorks housing 7 Hithium 1P52S modules.
- Laid out electrical schematics; 3D-printed prototypes; created fabrication drawings for mass CNC and waterjet fabrication

***Research Intern*, ModLab, University of Pennsylvania GRASP Lab** **Sept 2026 – Present**
- Project hardware lead designing and building a vortex-based wall-climbing robot to assemble plating on ships.

***Mechanical Subteam Member*, PennAiR Aerial Robotics, University of Pennsylvania** **Sept 2026 – Present**
- Responsible for the design of the entire micro-class aircraft in CAD for the 2027 SAE Aero Design competition; validated the design with XFLR5 and CFD simulations to maximize payload and minimize drag for a higher competition score.
- Fabrication using CNC, evaluated the aircraft and material properties including balsa, plywood, plastic and polycarbonate.

***Founder & CEO*, Enviro-AI (SPIRo)** **June 2023 – Present**
- Designed SPIRo, a modular pneumatic soft robot with ML gas-leak detection (95% accuracy).
- Founded VC-backed start-up; won the 1517 Fund Medici Grant; filed patent (published Jan 2026).

***Research Intern*, Chortos Lab, Purdue University** **June 2024 – June 2026**
- Developed 3D-printed HASEL soft actuators with integrated piezoresistive pressure sensing for cardiac assist devices.
- Integrated ML control and FEM/CFD modeling to optimize actuator response, improving prediction accuracy by 60%.

***Summer Research Intern*, The Faboratory, Yale University** **July – Aug 2025**
- Built a shape-morphing amphibious robot with variable-stiffness limbs for adaptive land and water locomotion.
- Proposed a deep-water robotic turtle and designed high-pressure waterproof motor casings rated to 10 m.

***Engineering Research Intern*, Schnitzer Lab, Stanford University** **Dec 2022 – June 2024**
- Designed circuit schematics and PCBs for sensing and control of experimental apparatus used across the lab.
- Implemented a high-throughput NAS storage system to store imaging data streaming at ~40 MB/s.

SELECTED PUBLICATIONS & PATENTS

- **Evan Zhang**, Alex Chortos. 3D-Printed HASEL Actuators With Integrated Fluidic Piezoresistive Pressure Sensor for Cardiac Assist Applications. *ICMSR 2025* (Dec 2025).
- **Evan Zhang**, Eddie Zhang. Omnimobile and Load Capable Pneumatic Soft Robot for Gas Distribution Pipeline Inspection. *IEEE RoboSoft 2025* (Apr 2025), Full Oral Talk.
- Eddie Zhang, **Evan Zhang**. Gas Pipeline Leakage Detection Based on Multiple Multimodal Deep Feature Selections and Optimized Deep Forest Classifier. *Frontiers in Environmental Science* (Apr 2025).
- **US Patent 2026/0008176 A1 (No. 19/274,609):** SPIRo: An AI Based Origami Inspired Pneumatic Soft Robot with Multidimensional Locomotion and Multimodal Deep Feature Selection and Fusion with an Improved Deep Forest Classifier Architecture for Real-Time Gas Pipeline Leak Detection (Jan 2026). **Evan Zhang**, Eddie Zhang

HONORS & AWARDS

- **Intl. Science & Engineering Fair (ISEF):** 2-Time Finalist and 2-Time Grand Prize 4th Place Winner (2025, 2026)
- **Regeneron Science Talent Search (STS)** Top 300 Scholar (Jan 2026)
- **Conrad Challenge:** Pete Conrad Scholar, 1st Place in Energy and Environment (Apr 2025)
- **Coca-Cola Scholars:** Semi-Finalist, one of ~1,500 selected from 100,000+ applicants (2025)

TEAMS & LEADERSHIP

***Policy Consulting Committee*, Penn Climate Ventures** **Sept 2026 – Present**
- Analyzing PJM grid voting records and stakeholder influence for NRDC to advance storage and renewable policy.

***Project Consultant*, Wharton Undergraduate Aerospace Club** **Sept 2026 – Present**
- Consulting for Angel Flight, a volunteer-pilot medical flight non-profit, to develop its national pilot scheduling system.

***Founder & President*, Harker 3D Printing and Design Club** **June 2024 – June 2026**
- Founded and scaled the club to 60+ members; taught CAD workshops and trained 60+ members to operate 3D printers.

***Co-President*, Harker Aerospace Club** **Aug 2022 – June 2026**
- Founded and led 2 teams in the American Rocketry Challenge; grew the club to 70+ members.

***Mechanical & Electrical Director*, FTC Team 11311 / Design Team, Harker FRC Team 1072** **Aug 2021 – Aug 2024**

SKILLS

- **Design & Controls:** SolidWorks, Onshape, KiCad, 3D printing, CNC, ROS, Arduino, myRIO, LabVIEW, NVIDIA Jetson
- **Simulation & Programming:** Abaqus, Ansys, Python, Java, C, MATLAB, PyTorch, scikit-learn, OpenCV, Verilog